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Prediction of dynamic strength index (DSI) using one low-cost 2D video camera: a machine learning approach

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This study examined whether a low-cost, two-dimensional (2D) video camera combined with supervised machine learning (ML) models could accurately predict the Dynamic Strength Index (DSI), a key indicator of neuromuscular performance. A total of 263 healthy participants performed countermovement jumps (CMJ) and isometric mid-thigh pulls (IMTP). Unlike the conventional method where both ballistic and isometric peak forces are measured using a force plate, in this study the isometric force was assessed using a back and leg dynamometer, while ballistic force during CMJ was estimated from 2D video data. Six spatiotemporal features extracted from the video, together with height and weight, were used to train multiple supervised regression models. Ground truth validation of the video-based ballistic force estimation was performed using force plate data. Gaussian Process Regression and Neural Networks achieved the highest prediction accuracy ($R^2 > 0.92$, RMSE < 0.06), demonstrating excellent agreement with laboratory-derived values. These findings highlight the potential of combining simple video systems with ML algorithms to estimate DSI accurately and affordably. The proposed method provides a scalable and non-invasive alternative to lab-grade equipment, enabling broader access to neuromuscular diagnostics and performance monitoring, particularly in youth, community sports, and field-based environments.

Keywords Dynamic strength index (DSI), 2D video analysis, Machine learning, Sports performance, Ground reaction force (GRF), Force plate

The integration of biomechanics in sports science has seen substantial growth in recent years, contributing significantly to improvements in athletic performance and injury prevention. Key fitness indicators such as jump height[1], the Reactive Strength Index[2], Rate of Force Development[3], and the Dynamic Strength Index (DSI)[4], alongside various assessment methods like the Countermovement Jump (CMJ), Drop Jump, and Isometric Mid-Thigh Pull (IMTP), have become pivotal tools in evaluating an athlete's physical readiness[5,6]. Among these, the DSI is particularly valuable as it offers a comprehensive measure of an athlete's ability to generate force dynamically, offering insights into both strength and ballistic force output during movement[4].

The DSI is a critical metric in sports biomechanics, often used to assess an athlete's neuromuscular capacity and their ability to produce explosive force in dynamic movements like jumps. It is calculated as the ratio of peak force during a ballistic movement (e.g., countermovement jump) to peak force during an isometric test (e.g., isometric mid-thigh pull), as illustrated in Eq. 1. This index provides valuable insights into the balance between strength and ballistic capabilities, which is crucial for optimizing athletic performance and minimizing injury risk[4,7,8]. By analyzing these forces, coaches and sports scientists can determine whether an athlete requires maximal strength training, ballistic strength training, or a concurrent training approach[9,10]. DSI can also be categorized to reflect strength–power profiles. Athletes with low DSI values (< 0.6) are often considered strength-dominant, while those with higher values (> 0.8) are power-dominant. These categories have practical significance in guiding individualized training and rehabilitation strategies, and thus served as one of the prediction outcomes in this study.

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While force plates remain the gold standard for measuring ground reaction forces (GRF) during these dynamic movements[11], their high cost and the logistical challenges of using them in real-world settings, such as sports training environments, limit their widespread application.

To address these challenges, recent research has increasingly focused on the development of low-cost, non-invasive tools for biomechanical assessment. Among these, 2D video cameras offer a promising alternative for capturing motion data that can be analyzed to predict various biomechanical parameters, including the DSI[12]. The use of standard video cameras in sports science has gained attention due to their affordability, accessibility, and the rapid advancements in image processing and computer vision technologies[13]. While earlier studies have demonstrated the feasibility of using cameras to estimate kinematic parameters[14,15], the direct prediction of DSI using a low-cost 2D camera and machine learning remains underexplored.

Supervised machine learning models, including but not limited to Artificial Neural Networks (ANNs), have emerged as powerful tools for modeling complex relationships between input features and target outcomes in various domains, including biomechanics[16]. These algorithms can learn from large datasets and capture nonlinearities and interactions between variables, offering the potential to improve prediction accuracy beyond traditional regression-based methods[17,18]. Different regression models were considered to capture varying levels of complexity. Linear models served as a transparent baseline, Gaussian Process Regression (GPR) was chosen for its robustness in small-sample nonlinear modeling, and Neural Network Regression was included to model complex nonlinear interactions. These complementary approaches enabled a more comprehensive evaluation of predictive performance.

Machine learning techniques have been applied successfully in domains such as gait analysis, motion prediction, and injury risk classification[18,19]. Despite these advances, most existing studies rely on expensive equipment or complex experimental setups, limiting their practical application in routine sports training. Using a low-cost 2D video camera combined with machine learning models, including ANNs, regression trees, and support vector machines, could lower the cost and complexity of biomechanical assessments while providing actionable insights for coaches and athletes. This approach aligns with the growing demand for affordable, non-invasive technologies that can be used in non-laboratory settings, such as local sports clubs, fitness centers, and rehabilitation facilities[20,21].

The primary objective of this study was to develop and validate a methodology for predicting the DSI using a low-cost 2D video camera system in combination with multiple machine learning algorithms. By capturing video data during dynamic movements like jumping and using supervised learning models for analysis, this research seeks to demonstrate a proof-of-concept for an accessible and scalable solution, with the potential to be extended and validated in both elite and amateur athletic settings in future studies. This method has the potential to make biomechanical assessments more accessible to a broader range of athletes and trainers, thereby contributing to better-informed training decisions and injury prevention strategies.

Materials and methods

Overview

To calculate the DSI, the peak isometric force was first obtained through an IMTP test using a back and leg dynamometer. The peak ballistic force during jumping was then determined based on data collected from a force plate and a two-dimensional (2D) video camera while participants performed a full CMJ. Data from each device were analyzed independently. The force plate data were segmented into distinct jump phases to identify the maximum ground reaction force (GRF) prior to takeoff. Concurrently, the video recordings were processed to extract identifiable spatiotemporal features, such as the duration of the descending phase. These features, together with anthropometric variables, were used as input for a set of supervised machine learning algorithms, which were developed to estimate peak ground reaction force and ultimately compute the DSI (Fig. 1).

Participants

A total of 263 participants (163 males and 100 females) voluntarily took part in this study. The participants had a mean age of 16.1 ± 5.6 years, an average height of 158 ± 16.6 cm, and an average mass of 50.9 ± 18.7 kg. Inclusion criteria required participants to self-report engaging in moderate physical activity at least three times per week for 30 min or more, in accordance with the guidelines of the American College of Sports Medicine. Exclusion criteria included any lower limb injury within the past six months. All participants wore dark spandex clothing and appropriate athletic footwear. The study was approved by the Institutional Review Board of Yazd University (Approval Code: IR.YAZD.REC.1403.078), and all participants provided written informed consent prior to participation. Moreover, all experiments were performed in accordance with relevant guidelines and regulations.

Data collection

Prior to the jump tests, participants completed three trials of an IMTP using a back–leg–chest dynamometer (Yagami, Japan). In this test, participants adopted a standardized position with hips and knees slightly flexed, gripping a fixed bar at mid-thigh height. A maximal isometric effort was exerted for 3–5 s in each trial, with one-minute rest intervals between attempts. The highest recorded value among the three trials was selected as the peak isometric force. This type of dynamometer was chosen due to its lower cost and greater accessibility compared to force plates, offering a practical alternative for field-based strength assessment.

Following the IMTP test, each participant performed five CMJs. Jumps were executed in response to a verbal command ("Jump"), with participants keeping their hands on their hips, feet parallel, knees fully extended, and thighs in a neutral position. A trial was considered valid if the movement began only after the command, both feet left and returned to the force plate simultaneously, and hands remained on the hips throughout the movement. To ensure consistency and familiarity with the procedure, at least two practice trials were performed prior to formal data collection. A rest interval of one minute was provided between each repetition.

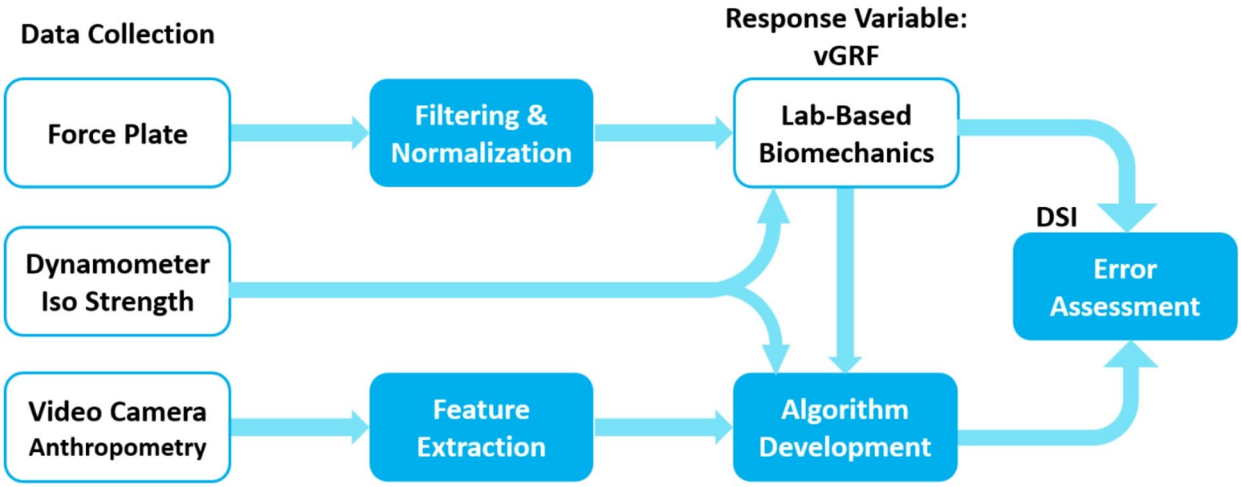


Fig. 1. Schematic overview of the project workflow, including data collection, biomechanical modeling, feature extraction, machine learning-based prediction, and validation of DSI estimates.

Vertical ground reaction force (vGRF) data were collected at a sampling frequency of 1000 Hz using a force plate (FP406020, Bertec Corp, Columbus, OH, USA). Simultaneously, a high-definition (HD) 2D video camera (Nikon D7100, Japan) with a commonly available 60 Hz frame rate was positioned 3 m to the right of the participant, perpendicular to the sagittal plane. The camera was mounted on a tripod at a height of 1 m, and the optical zoom was adjusted to ensure full capture of vertical motion, including a minimum field of view of 3 m.

To validate the 2D measurements, a motion capture system (OptiTrack Dou120, USA) was used concurrently. This system included two VGA-resolution cameras operating at 120 Hz. Three marker clusters were affixed to the posterior pelvis, mid-lateral thigh, and mid-lateral shank. Additional reflective markers were placed on anatomical landmarks, including the lateral femoral condyle, lateral malleolus, heel, and the distal end of the second metatarsal on the right foot (Fig. 2).

For the 2D analysis, only a single marker was attached to the pelvis. In order to convert pixel-based measurements from the video to real-world distances, a calibration frame with an L-shaped configuration and three non-collinear markers was mounted on the wall within the shared field of view of both the video and motion capture systems.

Laboratory-based Biomechanical analysis (force plate and camera)

To compute the DSI, two peak force values were required: the peak isometric force, obtained through the IMTP test using a back and leg dynamometer, and the peak ballistic force, extracted from the CMJ. DSI was computed as Ballistic Peak Force / Isometric Peak Force. For the dynamic component, the vertical ground reaction force (vGRF) prior to takeoff was identified using data from the force plate. These signals were filtered using a fourth-order low-pass Butterworth filter with a 100 Hz cutoff frequency and normalized to participants' body weight.

Kinematic data were extracted using Kinovea (version 2023.1), an open-source video analysis software that has been validated for measuring joint angles and displacement in sport and clinical contexts. This choice was made to balance cost-effectiveness and accuracy with a single low-cost camera, while avoiding the computational demands and reduced precision often reported with single-camera pose-estimation frameworks. From the 2D video recordings, six spatiotemporal features were extracted using Kinovea: duration of the descending phase, ascending phase, flight phase, takeoff phase, jump depth, and vertical displacement of the center of mass (COM). These features were obtained through a semi-automated process, based on a single marker placed on the pelvis. Instead of calculating maximum knee flexion angles, the lowest vertical position of the pelvic marker was used as the reference point for phase segmentation.

The six features were defined as follows (Fig. 3):

- The onset of the countermovement was defined as the first frame in which the pelvis marker showed a downward displacement relative to its baseline position, calculated as the mean vertical coordinate over the 200 ms pre-movement phase. This threshold corresponds to the initiation of hip and knee flexion and ankle dorsiflexion and is consistent with previously described methods in CMJ analysis.
- The **ascending phase** spanned from the lowest pelvis position to the moment of takeoff.
- The **takeoff phase** encompassed the entire movement from the start of descent to the instant both feet left the ground.
- The **flight phase** represented the duration from takeoff to landing.
- **Jump depth** was defined as the vertical displacement of the pelvis marker from the starting position to its lowest point.
- **COM displacement** was estimated as the vertical distance from the lowest pelvis position to the peak jump height, inferred from the apex of the flight phase.

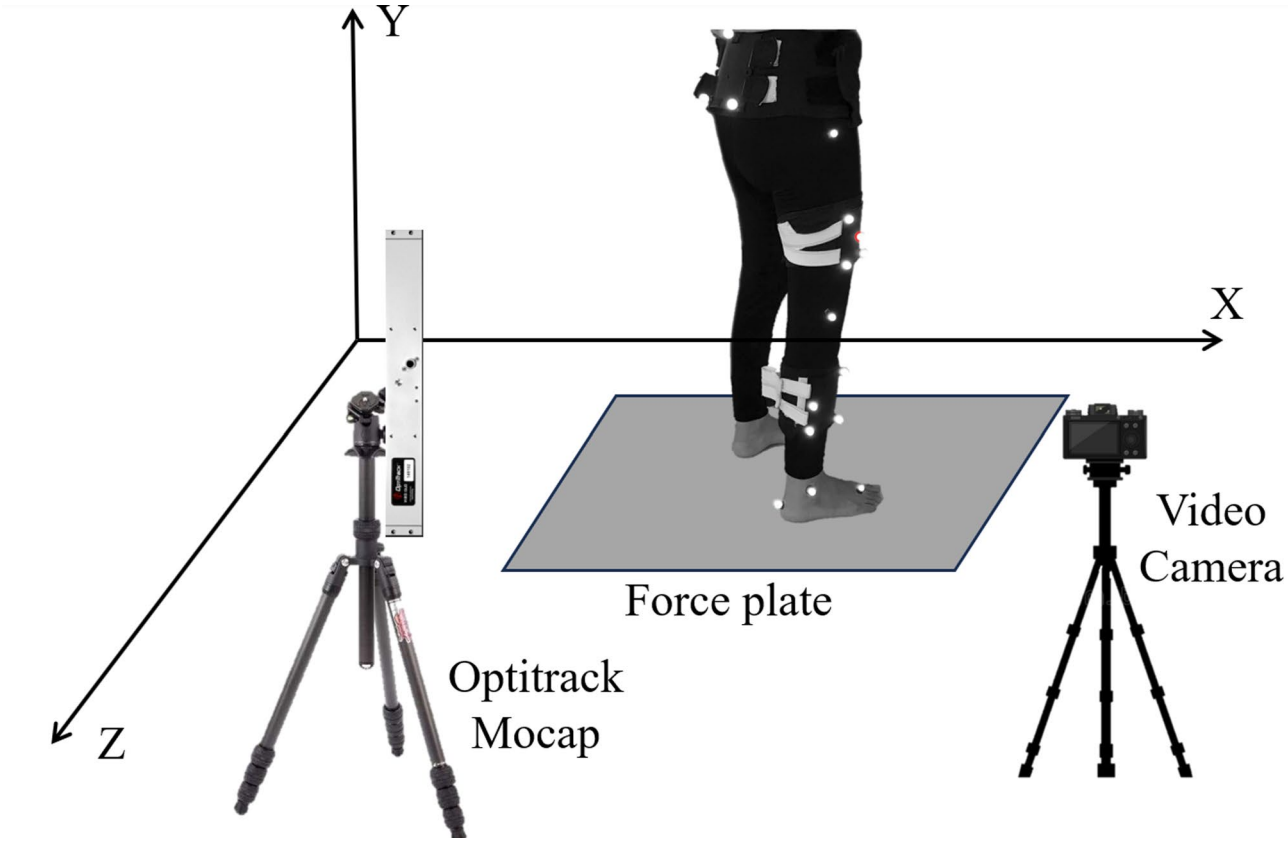


Fig. 2. Marker placement and system setup.

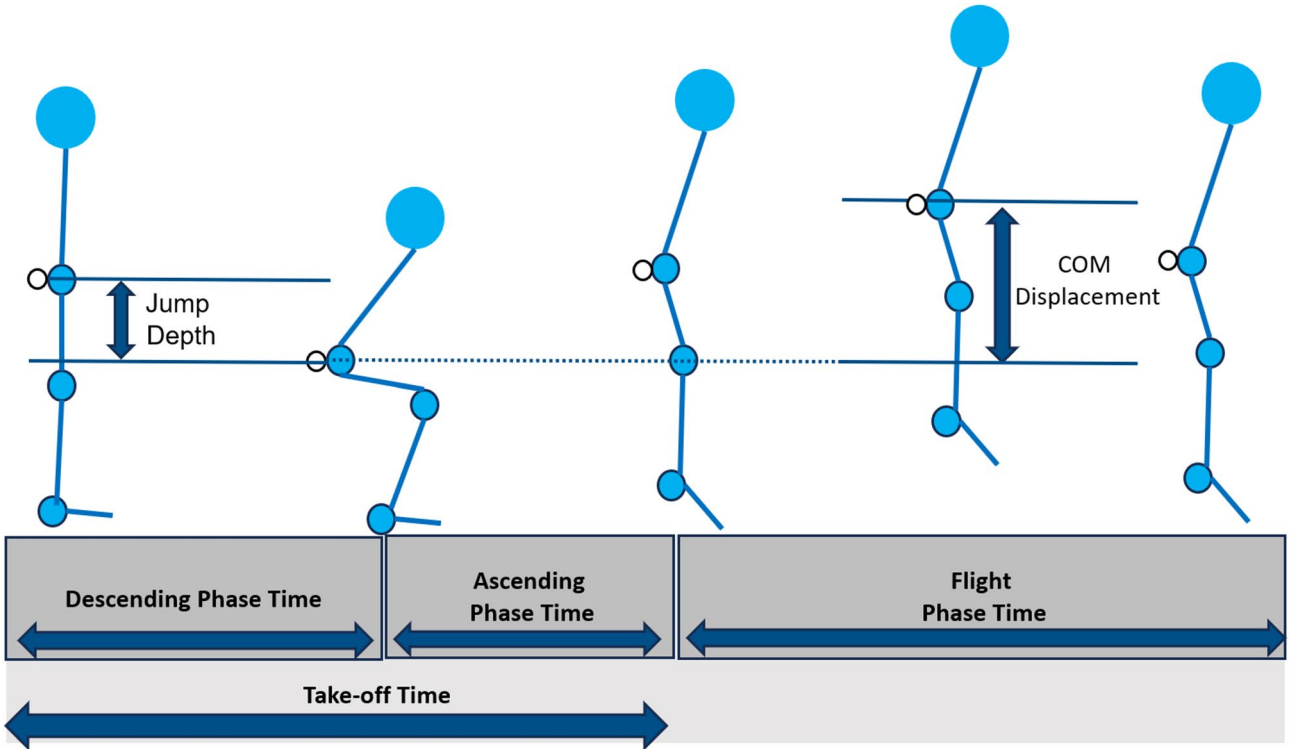


Fig. 3. Output components from the video camera system for algorithm input.

To validate these video-derived variables, a three-dimensional motion capture system was used. The center of the pelvis marker cluster was computed, and knee flexion angles were derived using segmental data from the thigh and shank through inverse kinematics. The timing of takeoff and landing was determined using foot markers. Although data from the force plate, video camera, and motion capture system were collected simultaneously, synchronization between systems was not required, as each dataset was processed independently.

Algorithm development

Following feature extraction, six spatiotemporal components related to the countermovement jump were combined with two anthropometric variables (height and weight), resulting in a total of eight input features. To enhance model stability, Principal Component Analysis (PCA) was employed for feature reduction. Although only eight features were initially extracted, PCA was applied to reduce potential collinearity and noise among predictors, thereby improving model stability and reducing the risk of overfitting. The first three components, which met the Kaiser criterion (eigenvalues > 1) and together explained more than 85% of the total variance, were retained for model training. Subsequently, model training and evaluation were conducted across four input scenarios:

- **Anthropometric features only** (height and weight; 2 features).
- **Spatiotemporal features only** (jump timing and displacement variables; 6 features).
- **Principal component features** (top 3 components from PCA applied to all 8 features).
- **All features combined** (full set of 8 features).

To predict ground reaction force and estimate DSI values, a set of supervised regression models was developed. These included linear regression, regression trees, support vector machines (SVM), Gaussian process regression (GPR), and neural network regression.

All algorithms were implemented using the Regression Learner Toolbox in MATLAB version 2021b. Hyperparameter optimization was performed using a grid search approach tailored to each model type. To enhance the reliability of the models and reduce variance due to data splitting, 5-fold cross-validation was applied. In all cases, 85% of the dataset was randomly assigned for model training and the remaining 15% was reserved as a common test set, which was consistently used across all models.

Algorithm evaluation

The performance of the developed regression models was evaluated using multiple statistical metrics. The coefficient of determination (R^2) and root mean square error (RMSE) were calculated to quantify the predictive accuracy of each model in estimating DSI values compared to those derived from laboratory measurements. Although no universal threshold for R^2 exists in exploratory biomechanics research, values above 0.80 were considered indicative of high prediction accuracy, consistent with previous studies using similar modeling approaches.

In addition to R^2 and RMSE, the normalized RMSE (nRMSE) was also calculated by dividing the RMSE by the standard deviation of the observed data, providing a scale-independent measure of error. The coefficient of variation (CV) was calculated to assess the relative dispersion of prediction errors. The intraclass correlation coefficient (ICC) was used to evaluate the consistency and agreement between the predicted and laboratory-derived DSI values.

Furthermore, Bland–Altman plots were used to visually inspect the agreement and potential systematic bias between methods. This combination of statistical and graphical analyses provided a comprehensive assessment of model validity across different input feature sets and regression approaches.

In addition to continuous regression evaluation, a classification-based approach was also employed to interpret model outputs in the context of training recommendations. As reported in the literature[2], three distinct DSI categories are commonly used. The first class includes DSI values less than 0.6, for which ballistic strength training and rate-of-force development exercises are typically recommended. The second class comprises values between 0.6 and 0.8, representing an optimal balance between ballistic and maximal strength, where concurrent training is suggested. The third class includes values greater than 0.8, indicating a relative deficiency in maximal strength capacity.

Due to the uneven distribution of samples across these categories, performance evaluation was based on Precision, Recall, and F1-score, which are widely accepted metrics for imbalanced classification problems. For each class, a one-vs-all strategy was used:

- In Class 1, predictions with DSI < 0.6 were labeled as 1, and all others as 0.
- In Class 2, predictions between 0.6 and 0.8 were labeled as 1, and all others as 0.
- In Class 3, predictions > 0.8 were labeled as 1, and all others as 0.

Precision, Recall, and F1-score were computed using the following formulas:

$$\text{Precision} = \text{TP} \,/\, (\text{TP} + \text{FP}) \tag{1}$$

$$\text{Recall} = \text{TP} \,/\, (\text{TP} + \text{FN}) \tag{2}$$

$$\text{F1} - \text{score} = (2 \times \text{Precision} \times \text{Recall}) \,/\, (\text{Precision} + \text{Recall}) \tag{3}$$

In these equations, *True Positive (TP)* refers to predictions correctly labeled as 1. *False Positive (FP)* refers to predictions incorrectly labeled as 1, and *False Negative (FN)* refers to predictions incorrectly labeled as 0.

	Features					
	Descending time (sec)	Ascending time (s)	Flight time (s)	Takeoff time (s)	Jump depth (cm)	COM displacement (cm)
R^2	0.99	0.96	0.97	0.99	0.99	0.99
RMSE	0.01	0.01	0.01	0.02	1.02	0.50
nRMSE	0.067	0.208	0.171	0.108	0.076	0.062
CV%	1.39	3.95	2.43	1.90	1.62	-1.77
ICC	0.99	0.98	0.99	`0.99	0.99	0.99

Table 1. Validity of spatiotemporal features extracted from 2D video analysis compared to reference values from 3D motion capture during countermovement jump. R^2, coefficient of determination; RMSE, root mean square error; CV, coefficient of variation; ICC, intraclass correlation coefficient

		Features			
		Anthropometric	Spatial-Temporal	Principal Component	All Features
	Features (#)	2	6	3	8
Linear Regression	R^2	0.81*	0.93*	0.86*	0.92*
	RMSE	0.11	0.06	0.09	0.06
	nRMSE	1.08	0.9	1.1	0.9
	CV%	14.48	7.91	11.88	8.02
	ICC	0.89	0.96	0.92	0.96
Regression Trees	R^2	0.88*	0.94*	0.85*	0.93*
	RMSE	0.07	0.05	0.08	0.06
	nRMSE	0.98	0.82	1.05	1.18
	CV%	9.22	7.09	11.62	7.34
	ICC	0.94	0.96	0.92	0.96
Support Vector Machine	R^2	0.83*	0.91*	0.88*	0.91*
	RMSE	0.09	0.06	0.07	0.07
	nRMSE	1.04	0.9	0.91	0.91
	CV%	11.8	8.12	9.7	9.41
	ICC	0.9	0.95	0.94	0.95
Gaussian Process Regression	R^2	0.9*	0.95*	0.91*	0.97*
	RMSE	0.07	0.05	0.06	0.04
	nRMSE	1.06	0.98	0.98	1.12
	CV%	9.33	6.31	8.59	5.11
	ICC	0.95	0.97	0.95	0.98
Neural Network Regression	R^2	0.85*	0.94*	0.85*	0.94*
	RMSE	0.08	0.05	0.08	0.05
	nRMSE	0.98	0.98	0.98	0.98
	CV%	11.35	7.03	10.78	6.7
	ICC	0.92	0.97	0.92	0.97

Table 2. Performance of supervised regression models in predicting the DSI across different input feature sets. R^2, coefficient of determination; RMSE, root mean square error; nRMSE, normalized root mean square error; CV, coefficient of variation; ICC, intraclass correlation coefficient. Values marked with an asterisk () indicate $R^2 > 0.80$, representing high prediction accuracy

Results

The agreement between the spatiotemporal features extracted from 2D video analysis and those obtained from the 3D motion capture system is summarized in Table 1. All six variables demonstrated good to excellent agreement, with intraclass correlation coefficients (ICC) ranging from 0.89 to 0.98. Coefficients of variation (CV) were consistently below 5%, which is considered indicative of excellent measurement consistency[22]. Additionally, root mean square errors (RMSE) were low across all features, further supporting the validity of the 2D video-based method for assessing countermovement jump performance.

The predictive performance of the regression models across different input feature sets is presented in Table 2. All models demonstrated moderate to high predictive accuracy, with several achieving R^2 values above 0.80 (indicated with an asterisk in the table). The highest performance was observed in the Gaussian process regression model, which consistently yielded R^2 values above 0.90 and CV values below 10% across all feature

combinations. Neural network and support vector machine models also performed well, while the linear regression model achieved comparable results in some conditions—reaching an R^2 of 0.92 when all features were included—suggesting that the relationship between predictors and DSI is not strongly nonlinear. Normalized root mean square error (nRMSE) values ranged from 0.82 to 1.18 across models. These error levels are generally considered acceptable in applied biomechanical research, supporting the feasibility of using these models for practical DSI estimation.

The agreement between predicted and laboratory-measured DSI values was further examined using Bland–Altman plots, presented in Fig. 4. For each regression model, the plots display the mean difference and 95% limits of agreement between predicted and actual DSI scores. The limits of agreement were relatively narrow across all models, and no clear systematic bias was observed. The distribution of residuals appeared random and centered around zero, indicating good agreement between predicted and measured values.

Figure 4 about here.

The classification results across DSI categories are presented in Table 3. The highest F1-scores were observed in Class 1 (DSI < 0.6) and Class 3 (DSI > 0.8), with Gaussian process regression and neural network models achieving values above 95% across most input conditions. In contrast, lower accuracy was seen in Class 2 ($0.6 \leq DSI \leq 0.8$), where F1-scores ranged from 59% to 91%. This decline is likely due to overlap between adjacent categories, making boundaries less distinct.

Among all models, Gaussian process regression consistently yielded the best performance across all classes, particularly when using all features. Neural networks also performed well, especially in the extreme categories. Linear regression, while effective for Classes 1 and 3, showed reduced accuracy in Class 2.

Discussion

This study demonstrated that the Dynamic Strength Index (DSI) can be predicted with good accuracy using kinematic features extracted from a single low-cost 2D video camera. The video-derived measures showed

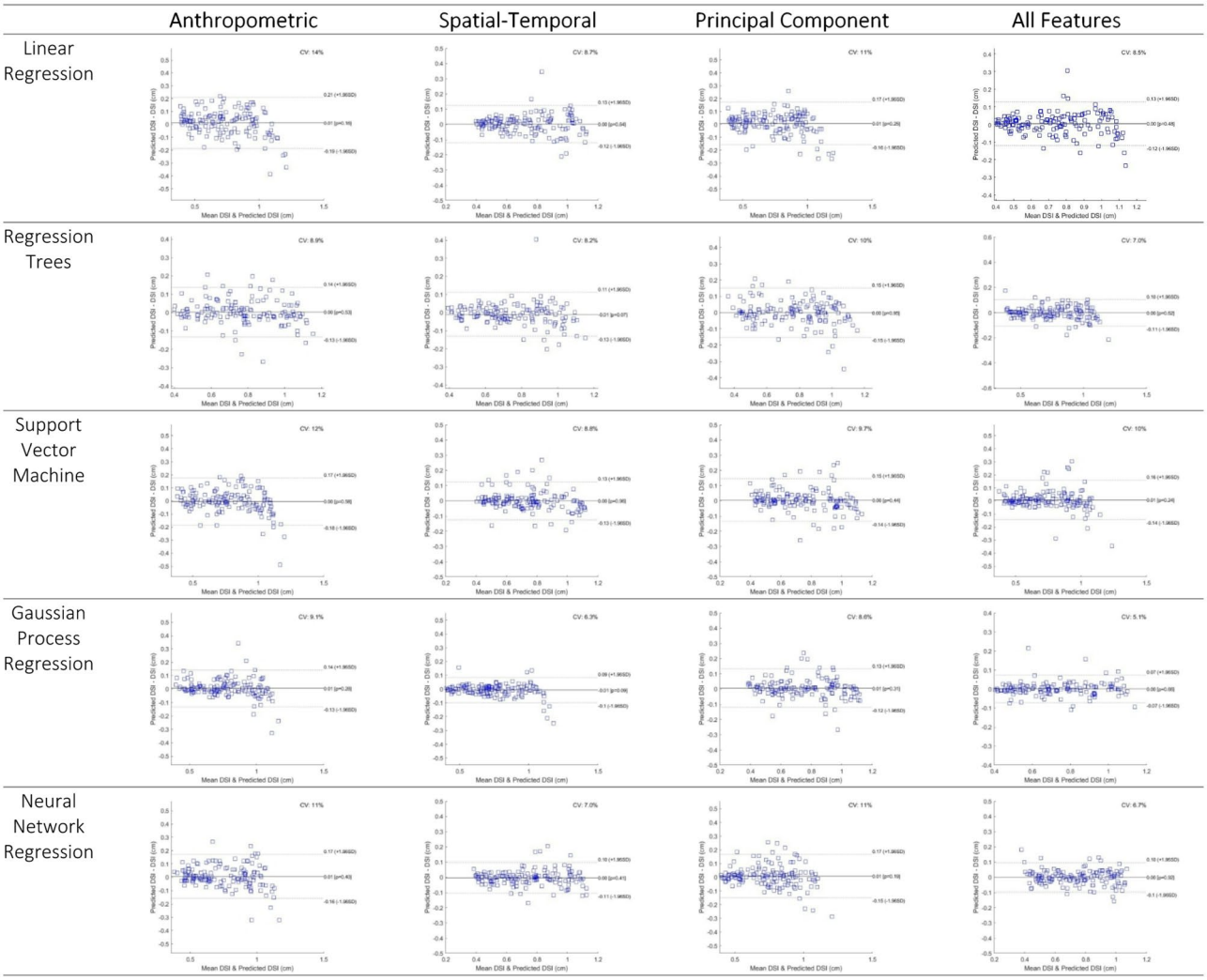


Fig. 4. Bland–Altman plots comparing predicted and measured DSI values across five regression models (Linear Regression, Regression Trees, SVM, GPR, and Neural Network) using different feature sets.

		DSI < 0.6			DSI ≤ 0.8, DSI ≥ 0.6			DSI > 0.8		
		Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)	Precision (%)	Recall (%)	F1-Score (%)
Linear Regression	Anthropometric	90.2	78.7	84.1	56.8	61.8	59.2	86.2	90.3	88.2
	Spatial-Temporal	97.8	93.8	95.7	89.7	83.9	86.7	94.1	100	97
	Principal Component	95.7	83	88.9	61.5	77.4	68.6	91.4	89.8	90.6
	All Features	93.2	87.2	90.1	74.4	71.1	72.7	82.1	90.2	86
Regression Trees	Anthropometric	91.9	82.9	87.2	81	79.1	80	90.6	98.3	94.3
	Spatial-Temporal	97.6	95.2	96.4	77.5	81.6	79.5	90.3	88.9	89.6
	Principal Component	95.8	83.6	89.3	63.4	66.7	65	79.6	87.8	83.5
	All Features	97	97	97	91.9	87.2	89.5	94.5	97.2	95.8
Support Vector Machine	Anthropometric	87.1	77.1	81.8	68.9	75.6	72.1	91	91	91
	Spatial-Temporal	90	92.3	91.1	81	75.6	78.2	88.5	91.5	90
	Principal Component	93.2	89.1	91.1	78.4	76.3	77.3	88.7	93.2	90.9
	All Features	97.7	87.8	92.5	76.9	78.9	77.9	88.3	94.6	91.4
Gaussian Process Regression	Anthropometric	95.7	95.7	95.7	87.8	83.7	85.7	91.1	94.4	92.7
	Spatial-Temporal	91.9	94.4	93.2	92.3	87.3	89.7	92.6	96.2	94.3
	Principal Component	85.4	83.3	84.3	78.4	69	73.4	89.2	98.3	93.5
	All Features	100	93.5	96.6	89.8	93.6	91.7	94.1	96	95
Neural Network Regression	Anthropometric	95.5	87.5	91.3	70.3	83.9	76.5	93.5	90.6	92.1
	Spatial-Temporal	95.2	95.2	95.2	84.8	86.7	85.7	92.7	91.1	91.9
	Principal Component	100	88.4	93.8	75	78.6	76.7	85.2	89.7	87.4
	All Features	100	91.5	95.6	79.1	91.9	85	94.7	91.5	93.1

Table 3. Precision, recall, and F1-score for classification of DSI categories across regression models.

excellent reliability (ICC > 0.98), confirming their suitability as input variables for predictive modeling. Among the tested regression models, Gaussian Process Regression (GPR) and Neural Network Regression achieved the highest accuracy, reflecting their superior ability to capture nonlinear relationships in the data. Importantly, the models were also able to predict DSI categories, which provide practical guidance for tailoring training strategies—identifying strength-dominant athletes who may benefit from ballistic training and power-dominant athletes who may require strength-oriented interventions. Together, these findings support the feasibility of computer-vision–based approaches as accessible, cost-effective tools for performance monitoring and individualized training design.

Both RMSE and normalized RMSE (nRMSE) values confirmed that the prediction error was low relative to the scale of the data. The inclusion of nRMSE improves interpretability by allowing comparisons across outcomes of different magnitudes, thereby strengthening the applied value of the findings. Furthermore, ICC values above 0.98 demonstrated excellent measurement reliability, consistent with previous reports that highlight the robustness of video-derived kinematics for applied biomechanics[23].

Among the tested approaches, both GPR and Neural Network Regression achieved the lowest error values. While neural networks typically excel in large and complex datasets due to their capacity to model highly nonlinear interactions, the relatively small sample size and reduced feature space (after PCA) in the present study favored the stability of GPR. By employing kernel-based learning, GPR was able to model nonlinear patterns efficiently without requiring extensive parameterization, thereby reducing susceptibility to overfitting. This likely explains why GPR matched or even outperformed Neural Networks in certain scenarios, a finding consistent with prior reports of its robustness in small-to-moderate biomechanics datasets[17,18,24]. Interestingly, the linear regression model also showed comparable accuracy in some conditions, suggesting that the relationship between input features and DSI retains a partially linear structure.

This study extends the work of Lees et al. (2004) and Pueo (2016), who demonstrated the validity of 2D video data for kinematic analysis, by being among the first to apply machine learning for direct prediction of DSI from such data[25,26]. The excellent agreement observed between 2D video-derived kinematic parameters and motion capture ($R^2 \geq 0.96$ for all features) supports earlier findings by Monnet et al. (2014) regarding the reliability of non-laboratory devices for jump performance evaluation[23]. While most previous studies have relied on force platforms and 3D motion capture for DSI or related measures[27,28], these tools are expensive, immobile, and not widely accessible. The current approach offers a practical, scalable solution that aligns with the growing movement toward low-cost, field-based sports technologies[29].

The study's finding that using all eight input features (anthropometric + spatiotemporal) resulted in the highest prediction accuracy corroborates the importance of multimodal input integration, as emphasized by Camomilla et al. (2018), who found that combined sensor data enhances model robustness in real-world biomechanical applications[29]. Moreover, the feature reduction via Principal Component Analysis (PCA) retained competitive performance, indicating potential for further optimization of input parameters, especially in constrained data collection environments.

From a practical perspective, the successful use of a 60 Hz consumer-grade camera reinforces the trend toward accessible sports science tools. This aligns with the broader movement in wearable and computer vision-based sports technologies aiming to bring advanced analytics to grassroots and rehabilitation settings[30].

The algorithms developed in this study were implemented within the graphical user interface of MATLAB's Regression Learner App (version 2021b), enabling straightforward model training and DSI prediction. Although a dedicated mobile application is not currently available, the underlying regression equations and implementation code can be shared upon request. These models are compatible with widely used platforms such as MATLAB and Python, facilitating adoption in various field and clinical settings.

This study has several limitations. First, the sample consisted of adolescents and young adults, which restricts immediate generalizability to elite athletes, older adults, or clinical populations. Second, testing was conducted under controlled laboratory conditions, limiting ecological validity. Third, the approach relied on 2D video and semi-automated feature extraction using Kinovea, which, while cost-effective, may be less precise than multi-camera markerless systems. Another limitation is the absence of formal model interpretability analyses (e.g., SHAP values or variable importance rankings). Although the present focus was on establishing feasibility and predictive accuracy, future work should integrate explainability frameworks to identify the most influential predictors, thereby improving practitioner trust.

Future research should expand validation to diverse cohorts and real-world environments, including sport-specific tasks such as sprinting and change-of-direction maneuvers. Automation pipelines using pose-estimation frameworks (e.g., OpenPose, MediaPipe) should also be systematically evaluated to test their feasibility for real-time deployment. In addition, hybrid model architectures that integrate biomechanical constraints with machine learning could improve generalization, and the combination of video features with other low-cost sensors (e.g., IMUs) may further enhance predictive performance without compromising accessibility. Finally, deployment via user-friendly platforms such as Python/Streamlit dashboards or mobile applications will be essential to maximize translation into both sports and clinical practice.

In conclusion, this study provides a proof-of-concept that DSI can be accurately predicted from simple video-derived kinematic features using machine learning models. The excellent reliability of input features, superior performance of nonlinear regression models, and ability to classify athletes into meaningful DSI categories highlight the applied potential of this approach. These findings support the broader use of cost-effective, computer-vision–based methods for individualized training design and rehabilitation planning. Future research across more diverse populations and ecological settings will be critical to establish generalizability and maximize real-world impact.

Data availability

The datasets generated and analyzed during the current study, including the independent validation dataset, have been deposited in the Zenodo repository and are publicly available at [https://doi.org/10.5281/zenodo.17153984].

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Author contributions

Author 1 (M.H.L): Conceptualization, Data Curation, Investigation, Methodology, Writing – Original Draft. M.H.L. developed the research plan and methodology, conducted the statistical analysis, and drafted the initial version of the manuscript. **Author 2 (A.A):** Conceptualization, Investigation, Writing – Review & Editing. A.A. curated the data and provided critical revisions and feedback on the manuscript.

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Declarations

Competing interests

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